

DarkAR: Augmented Reality Expression Method in Dimly lit Environment

Masami Hirabayashi
Institute of Advanced Media Arts and
Sciences
Ogaki, Gifu, Japan
hrr@iamas.ac.jp

Kakuya Shiraishi
Tokyo Polytechnic University
Nakano, Tokyo, Japan
kakuya.sris@gmail.com

Scott Allen
Kyoto Seika University
Kyoto, Kyoto, Japan
scottallen@kyoto-seika.ac.jp

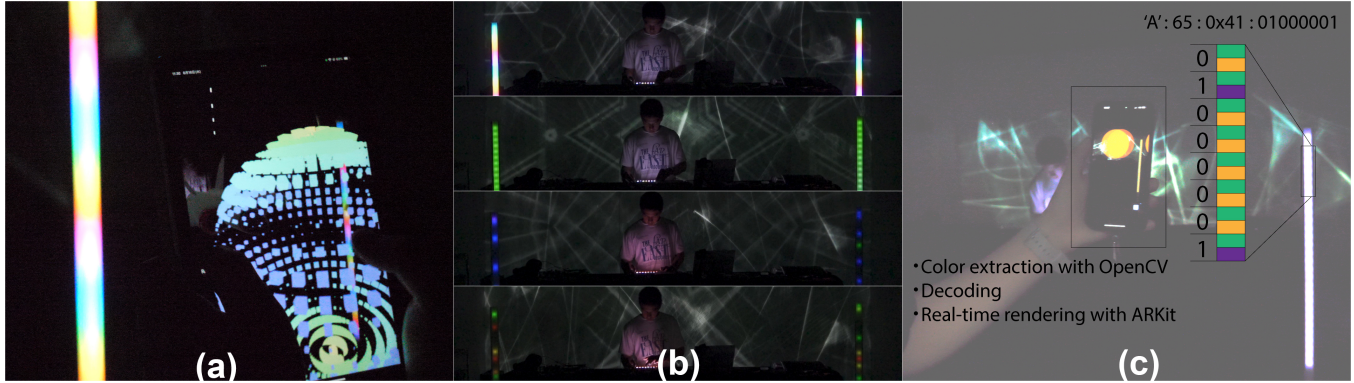


Figure 1: DarkAR Expressions and System

Abstract

We propose a method for augmented reality (AR) expression in dimly lit spaces. Although AR superimposed on the real world has become a popular form of expression, AR expression using smartphones and HMDs is difficult in dimly lit (dark) places because it is impossible to grasp the space. Music venues, in particular, are often dimly lit, which often renders AR ineffective. Our method utilizes LED lighting devices used for staging in music venues to create an AR expression method that can be easily used by the audience. Using light transmission from flickering full-color LEDs as a substitute for AR markers makes it possible to blend AR expression into the space such as music venues without ruining atmosphere. This technique is readily available to anyone who can realize it with a smartphone alone, and is one method that opens up AR expression in dimly lit spaces.

Keywords

AR, Dark Place, Music Venue, Optical Communication

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1 Introduction

In Smartphones, video-passthrough-HMD and AR glasses, XR has recently become more popular for expression, especially after the onset of the pandemic. At NxPC.Lab [NxPC.Lab 2025], our research platform for augmenting musical experiences, we have also explored the potential of XR representation. However, AR using smartphones or HMD video pass-through usually does not work in dimly lit places and is often unavailable in music venues, which are the subject of our research. In this study, we adopted a methodology that enables AR expression in music venues and can easily be used while enjoying music, without requiring special procedures.

2 Concept & Objective

Our goal is to develop an AR method that can easily be used while enjoying music at venues, etc., without requiring the audience to prepare in any way. It can be used simply by launching an application and pointing a camera at the performer. Therefore, it must not require WiFi or BLE and be realizable with a smartphone or HMD alone. Complicated processes discourage audience participation. Additionally, music venues, which are often located underground, have difficulty providing stable internet access when accessed by a large number of people. Most importantly, audience members should be able to enjoy the music naturally without disrupting the venue's atmosphere. Therefore, we propose using a lighting device (in this case, a staging device with LED strips) to replace the AR markers. This device can be installed in the venue without appearing incongruous. When placed on the stage, this lighting device can superimpose AR expressions as if they were naturally surrounding the stage, simply by having the audience point their cameras at the stage. Thus, by starting the application and pointing it at the stage,

the audience can easily experience AR expression. Simply point a smartphone at the lighting device. We believe that this method of enabling AR expression in dark places has the potential to expand the possibilities of AR expression in dimly lit places.

3 Related Works

Live performances using AR include Gorillaz "Skinny Ape" Live Performance and CHAUSIE, which superimpose AR in conjunction with camera movements [Inc. 2020], as well as those using markers. AR in dark spaces includes those using smartphones and infrared cameras, such as InfraredTangs [Dogan et al. [n. d.]], and SLAM with AI-based illumination compensation [Savinykh et al. [n. d.]] [Singh et al. [n. d.]]. However, the only AR realized with smartphones or HMDs alone is gesture recognition using LiDAR, and almost none of them recognized space. Many studies have been conducted on how to use smart phones for optical communication as Visible Camera Communication. PicaPicamera [Iizuka and Kikuchi 2014] is another example of a method that uses the color of light.

4 System Overview

The system consists of a full-color LED strip, a microcontroller M5Stack, and a smartphone (iPhone). An overview of the system is shown in Figure 1(c). The 1.3-meter-long LED pole has a 1-meter-long Adafruit Dotstar full-color LED strip (with a maximum data rate of 32 MHz) attached to an aluminum frame and a translucent white acrylic plate to diffuse the light. The color and timing of the light emission are controlled by the M5Stack using Arduino. An example of LED poles placed on a stage is shown in Figure 1(b). The device can be installed in front of or beside the stage and used as a lighting device for stage effects at music venues. During installation, it can be controlled via Bluetooth to test brightness and adjust the lighting according to the venue. On smartphones (tested on iPhones 12 and above), the system uses ARKit and OpenCV with Swift on iOS. ARKit acquires 30 fps or 60 fps camera images, which are converted to the HSV color space to recognize the colors of the LED emissions. When an ID is detected in the LED emission color pattern using the predefined color pattern, the AR content is displayed in front of the camera. After the AR display, the display position in the AR space does not change because it changes according to the inertial control of the ARKit IMU. Since the AR display is performed by pointing the camera toward the stage and recognizing the LED emission pattern, the AR display will appear around the stage.

4.1 LED Transmission Encoding

The LED emission pattern uses three colors to implement multispectral communication, as shown in Figure 1(c). The color following the green color (orange or purple) is used to encode "0" and "1" to form an ID. Because smartphones use the ARKit framework, it is not possible to control the camera's exposure and other details. Therefore, when the camera acquires light from the LEDs emitted by the device in a dark place, the white light tends to stand out, making colors difficult to recognize. The camera uses an acrylic plate to diffuse the light and adjust the brightness of the LEDs. It also uses image processing to address this issue.

5 AR Expressions

This AR representation was created for NxPC.Live Vol. 73, held on March 19, 2025, at Circus Tokyo in Shibuya, Tokyo. The event was a club/DJ event. The design aimed to create a "sense of depth" and "spatial extension" to allow the audience to enjoy the event visually, even in the dimly lit environment of a music venue. The emphasis is on "visual elements that appear only in dark places" and "staging that blurs the boundary between reality and unreality." Specifically, we aim to create a new immersive experience by combining vision concealment and spatial reconstruction with a grid. In this experience, the audience temporarily forgets the real space, and sound, light, and images are integrated at the music venue. Simple, abstract black-and-white movements are visually harmonized with electric, rhythmic beats, creating an immersive effect. An example of an actual AR representation is shown in Figure 1(a).

6 Conclusion

We proposed "DarkAR," a system that enables AR expression in harmony with the environment and integrated with the performance, even in dimly lit places such as music venues.

This method allows for simple, smartphone-based AR production in dimly lit places and expands the possibilities of AR expression in such places. Currently, recognition may be unstable due to interference from the environment or VJ images, so we are working to improve the method. At the same time, we are planning to support video pass-through with HMDs. Additionally, to further explore AR expression in dimly lit places, we plan to expand into artistic and spatial expression, such as AR installations in dark places.

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